

EEE 310 (July 2023)

Communication Laboratory

Final Project Report

Section: **C1** Group: **05**

**EcoWatch: Smart Greenhouse Environment
Management with Fire Safety**

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1. Abstract

The emergence of the Internet of Things (IoT) has paved the way for remote monitoring and visualization of environmental parameters through wireless sensors, facilitating real-time data transmission and analysis. Our project, EcoWatch: Smart Greenhouse Environment Management with Fire Safety, leverages IoT principles to enhance the functionality and efficiency of traditional home appliances, focusing on greenhouse environment management and fire safety.

Using an Arduino UNO board and an ESP8266 Wi-Fi Module, we establish a network for processing and transferring data from sensors to the Thingspeak Cloud, a versatile open API service. The project integrates sensors such as the DHT11 for monitoring temperature, humidity, and soil moisture levels, enabling continuous tracking of environmental conditions at a cloud level. By assigning specific channel IDs and API keys, we ensure seamless data tracking at predefined intervals.

The primary objective of our project is to optimize the operation of home appliances by harnessing Arduino's adaptability. Through intelligent automation facilitated by sensors, actuators, and communication modules, we aim to enhance user experience and resource efficiency. The project encompasses design, implementation, and testing phases to ensure reliability and effectiveness.

In addition to environmental monitoring, our project addresses fire safety concerns, particularly in regions like Bangladesh where centralized fire detection and monitoring systems are lacking. We propose a comprehensive fire detection system that not only identifies the presence and location of fires but also alerts users via mobile notifications and stores data on a server for future reference. By integrating fire detection with our IoT framework, we aim to enhance safety measures and mitigate the risks associated with fire outbreaks in residential and commercial spaces.

Overall, EcoWatch represents a holistic approach to smart home management, encompassing environmental monitoring, resource optimization, and safety enhancement through the seamless integration of IoT technologies and traditional appliance systems.

2.Introduction:

The management of environmental conditions in controlled settings such as greenhouses poses a significant challenge due to the dynamic nature of factors like temperature, humidity, and soil moisture. Ensuring optimal conditions for plant growth while addressing concerns such as energy efficiency and fire safety requires a multifaceted approach. This project, EcoWatch: Smart Greenhouse Environment Management with Fire Safety, tackles these challenges by employing a combination of sensor technology, automation, and IoT principles.

The problem at hand is inherently complex, involving the coordination of various parameters to maintain an environment conducive to plant growth. Traditional methods of greenhouse management often rely on manual intervention, leading to inefficiencies and suboptimal

outcomes. Moreover, the risk of fire outbreaks in residential and commercial spaces adds another layer of complexity, necessitating proactive detection and mitigation measures.

Addressing these challenges requires an interdisciplinary approach that integrates engineering principles from fields such as electronics, software development, and environmental science. The design and implementation of an effective solution involve considerations such as sensor selection, actuator control algorithms, data communication protocols, and safety mechanisms.

Alternative solutions to greenhouse management and fire safety exist, ranging from manual monitoring and control systems to fully automated commercial solutions. However, these alternatives often lack the flexibility, scalability, and customization offered by a DIY IoT-based approach. Additionally, while some fire detection systems focus solely on alerting occupants, our project aims to enhance safety measures by integrating fire detection with environmental monitoring and data logging capabilities.

By addressing the complexities inherent in greenhouse management and fire safety through an integrated IoT framework, EcoWatch seeks to provide a comprehensive solution that improves efficiency, sustainability, and safety in controlled environments.

3.Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The project aims to address the challenges in environmental monitoring and safety systems within greenhouse agriculture, with a focus on integrating advanced sensor technologies and intelligent control systems to optimize resource utilization and ensure plant health.

3.1.2 Literature Review

Extensive research will be conducted to review existing literature, studies, and technological advancements in the fields of environmental monitoring, sensor integration, and agricultural automation. This will provide insights into best practices, emerging trends, and innovative solutions relevant to the project objectives.

3.1.3 Formulation of Problem

Based on the identified scope and insights from the literature review, specific engineering problems will be formulated, such as optimizing temperature and humidity control, automating irrigation based on soil moisture levels, and implementing fire detection and suppression

systems. These problems will be defined clearly to guide the project's objectives and design considerations.

3.1.4 Analysis

The project will involve the analysis of complex electrical and electronic engineering problems using principles of mathematical, natural, and engineering sciences. Data collected from sensors will be interpreted and analyzed to make informed decisions regarding environmental conditions, plant health, and safety measures. Advanced analytical techniques, including machine learning algorithms and predictive analytics, may be employed to enhance the accuracy and efficiency of the analysis process.

3.2 Design Method (PO(a))

The design methodology employed for this project encompasses several key components aimed at ensuring the successful implementation of an integrated environmental monitoring and safety system for greenhouse agriculture. Beginning with an understanding of the project requirements and objectives, the design process leverages various hardware components such as the ESP8266 Arduino module, DHT11 sensor, LCD display, and Arduino Uno board. These components are carefully selected based on their compatibility, functionality, and suitability for the intended application.

The design process involves thorough research and analysis of available technologies and platforms, including ThingSpeak, an IoT analytics platform, which facilitates data aggregation, visualization, and analysis in the cloud. By integrating ThingSpeak with the ESP8266 module, real-time sensor data streams can be transmitted and analyzed remotely, enabling efficient monitoring and control of greenhouse environmental parameters.

Furthermore, the design methodology incorporates rigorous testing and validation procedures to ensure the reliability and performance of the system. This includes prototyping and simulation using tools like Tinkercad to verify circuit functionality and identify potential issues before deployment. Through iterative testing and optimization, the design team validates the functionality of individual components and their interactions within the overall system architecture.

Additionally, attention is paid to electrical safety considerations, such as voltage regulation and component protection, to safeguard against potential damage or malfunction. This involves careful circuit design and testing to mitigate risks associated with overvoltage or other electrical hazards.

Overall, the design methodology emphasizes a systematic approach to problem-solving, drawing upon principles of electrical and electronic engineering to develop a robust and efficient solution tailored to the specific requirements of greenhouse agriculture.

3.3 Circuit Diagram

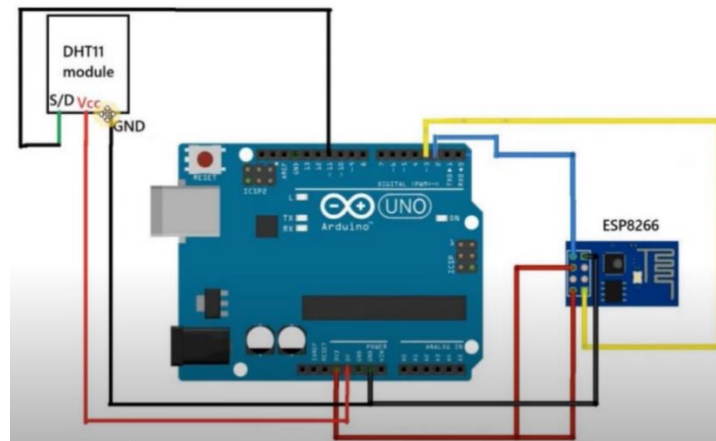


Fig: Arduino Uno to ESP8266 Connection

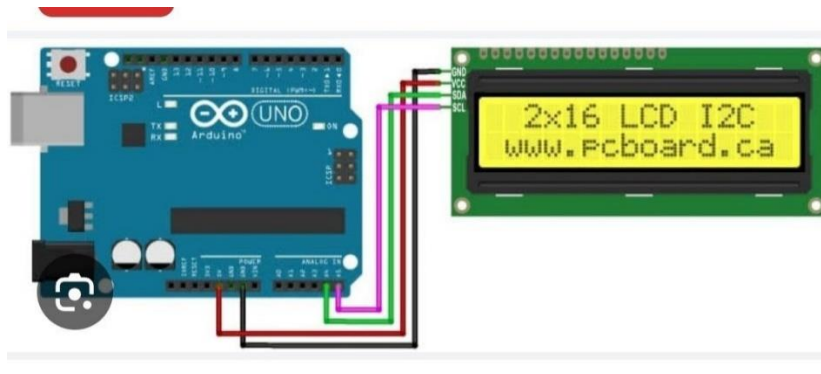


Fig: Arduino Uno to LCD Connection

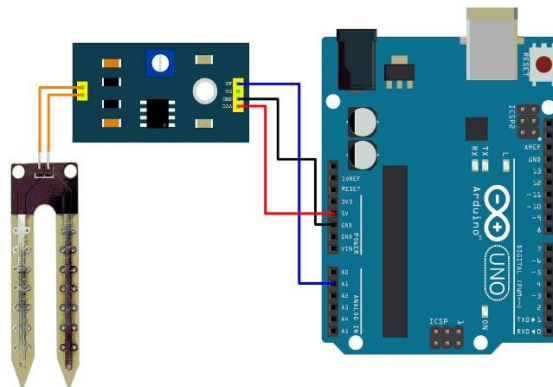


Fig: Arduino Uno to Soil Moisture Sensor Connection

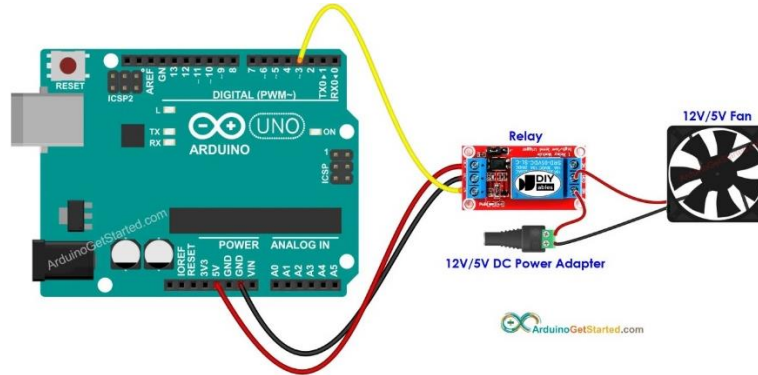


Fig: Relay to DC Fan Connection

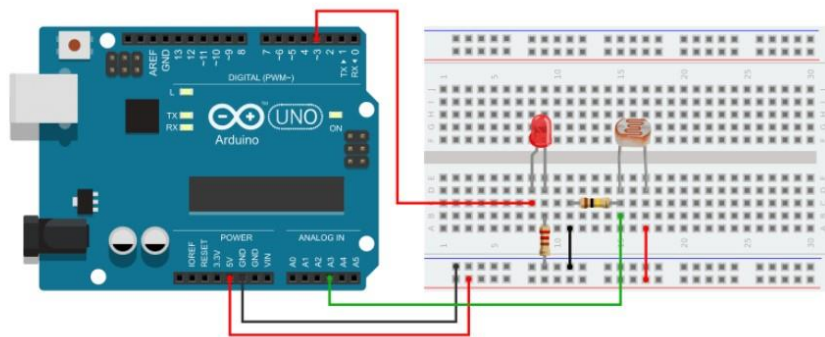


Fig: LED intensity control with LDR connection

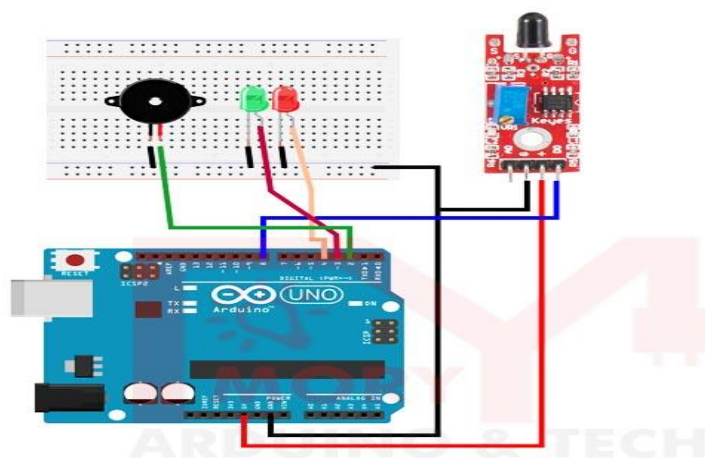


Fig: Flame sensor with buzzer connection

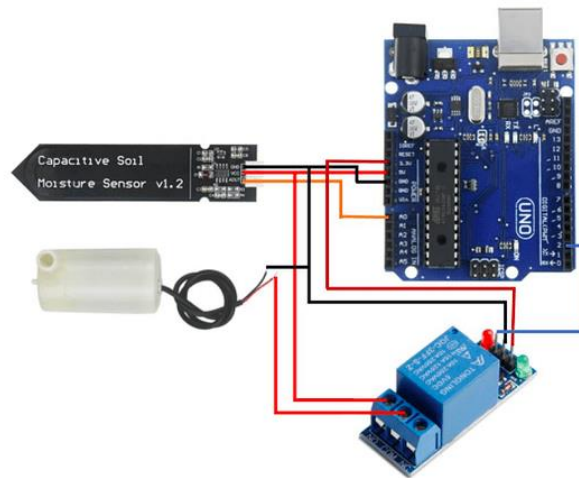
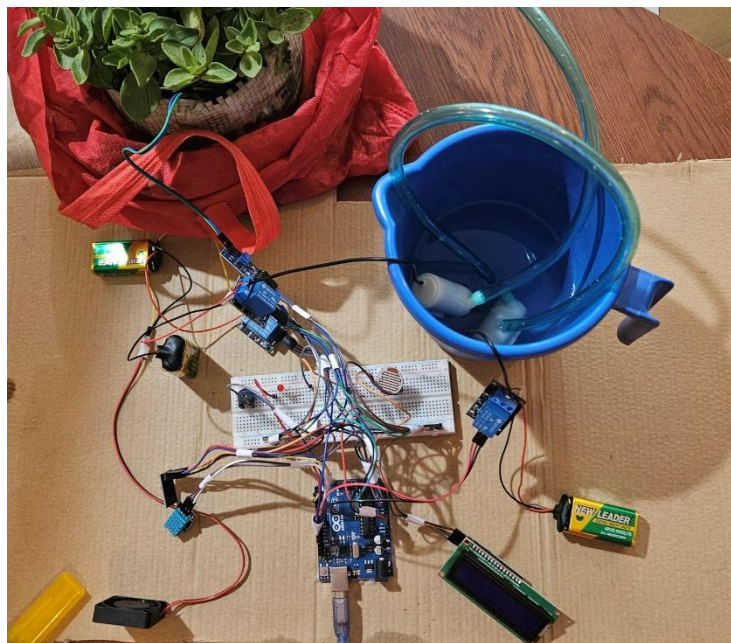


Fig: DC pump connection

Overall Model:



3.4 Full Source Code of Firmware:

```

// Check flame detection and control fire pump
flameSensorLoop();
// Control the soil pump based on the specified conditions
if (moisture_percentage < 20) {
  Serial.println("Pump turned on: Soil moisture below 20%.");
  digitalWrite(soil_pump_RELAY_PIN, HIGH); // Turn on the pump
  // Keep the pump on until the soil moisture is between 50% and 60%
  while (moisture_percentage < 50) {
    moisture_percentage = readSoilMoisture(); // Update soil moisture reading
  }
  digitalWrite(soil_pump_RELAY_PIN, LOW); // Turn off the pump
  Serial.println("Pump turned off.");
} else if (moisture_percentage > 65) {
  // Turn on the fan until soil moisture reaches 60%
  Serial.println("Fan turned on: Soil moisture above 65%.");
  digitalWrite(FAN_RELAY_PIN, HIGH); // Turn on the fan
  digitalWrite(10, HIGH); // Turn on the light

  // Keep the fan and light on until soil moisture is below 60%
  while (moisture_percentage > 60) {
    moisture_percentage = readSoilMoisture(); // Update soil moisture reading
    temperature = dhtObject.readTemperature(); // Update temperature reading

    // Check if temperature reaches 24 degrees Celsius and turn off the light
    if (temperature >= 24) {
      digitalWrite(10, LOW); // Turn off the light
      Serial.println("Light turned off: Temperature reached 24 degrees Celsius.");
    }
  }

  digitalWrite(FAN_RELAY_PIN, LOW); // Turn off the fan
  Serial.println("Fan turned off: Soil moisture below 60%.");
} else if (moisture_percentage > 20 && moisture_percentage < 60) {
  digitalWrite(soil_pump_RELAY_PIN, LOW);
  // Do nothing if soil moisture is between 20% and 60%
}

// Check flame detection and control fire pump
flameSensorLoop();

// Update ThingSpeak with sensor data
String getData = "GET /update?api_key=" + API + "&field3=" + String(temperature) + "&field2=" + String(humidity) +
"&field1=" + String(moisture_percentage);
sendCommand("AT+CIPMUX=1", 5, "OK");
sendCommand("AT+CIPSTART=0,\"TCP\", \"\" + HOST + "\",\" + PORT, 15, \"OK\");
sendCommand("AT+CIPSEND=0,\" + String(getData.length() + 4), 4, ">");
esp8266.println(getData);
delay(3000);
countTrueCommand++;
//sendCommand("AT+CIPCLOSE=0", 5, "OK");
}

float readSoilMoisture() {
  int sensor_analog = analogRead(soil_sensor_pin);
  float moisture_percentage = (100.0 - (sensor_analog / 10.23)); // Assuming soil sensor output is in the range 0-1023
  Serial.print("Soil Moisture Percentage: ");
  Serial.print(moisture_percentage);
  Serial.println("");
  return moisture_percentage;
}

void sendCommand(String command, int maxTime, char readReplay[]) {
  Serial.print(countTrueCommand);
  Serial.print(". at command => ");
  Serial.print(command);
  Serial.print(" ");
  while (countTimeCommand < (maxTime * 1)) {
    esp8266.println(command);
    if (esp8266.find(readReplay)) {
      found = true;
      break;
    }
    countTimeCommand++;
  }

  if (found == true) {
    Serial.println("OYI");
    countTrueCommand++;
    countTimeCommand = 0;
  }
}

```

```

}

if (found == false) {
    Serial.println("Fail");
    countTrueCommand = 0;
    countTimeCommand = 0;
}

found = false;
}

// Flame sensor code
void flameSensorLoop() {
    // Variables to track flame detection
    Serial.println("Checking if there is fire!");
    int flameCount = 0;
    const int flameThreshold = 600; // Adjust as needed (600 * 5ms = 3 seconds)
    const int requiredFlameReadings = 2; // Adjust as needed
    bool fireDetected = false;

    // Take multiple readings to debounce
    for (int i = 0; i < flameThreshold; i++) {
        if (digitalRead(flamePin) == LOW) {
            flameCount++;
        }
        delay(5); // Adjust as needed
    }

    // If flame detected in most of the readings, trigger fire relay
    if (flameCount >= requiredFlameReadings) {
        fireDetected = true;
    } else {
        fireDetected = false;
    }

    // Control the fire relay based on flame detection
    if (fireDetected) {
        Serial.println("Fire detected!");
        tone(buzzerPin, 1000); // Activate the buzzer at a frequency of 1000Hz
        digitalWrite(fire_pump_RELAY_PIN, HIGH); // Turn on the fire relay

        // Continuously check for fire until extinguished
        while (fireDetected) {
            // Check flame status
            flameCount = 0;
            for (int i = 0; i < flameThreshold; i++) {
                if (digitalRead(flamePin) == LOW) {
                    flameCount++;
                }
                delay(5); // Adjust as needed
            }
            // Update fire detection status
            if (flameCount >= requiredFlameReadings) {
                fireDetected = true;
            } else {
                fireDetected = false;
            }
        }

        // Fire extinguished, turn off the fire relay
        Serial.println("No fire detected.");
        digitalWrite(fire_pump_RELAY_PIN, LOW); // Turn off the fire relay
        noTone(buzzerPin); // Turn off the buzzer
    } else {
        // No fire detected, turn off the fire relay
        Serial.println("No fire detected.");
        digitalWrite(fire_pump_RELAY_PIN, LOW); // Turn off the fire relay
        noTone(buzzerPin); // Turn off the buzzer
    }
}
}

```

4 Implementation

4.1 Description

Components Used:

Microcontroller:

- Arduino Uno Board

Sensors:

- DHT11 sensor

Display:

- Liquid Crystal Display (LCD)

IoT Connectivity:

- ESP8266 module

IoT Platform:

- ThingSpeak

Other Components:

- Various passive components (resistors, capacitors, etc.)
- LED (used for testing)
- Breadboard
- Connecting wires

Thingspeak: ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualize, and analyze live data streams in the cloud.

ESP8266: ESP8266 Arduino compatible module has a Micro Controller Unit which gives the possibility to control I/O digital pins via simple code like programming language.

DHT11: DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature.

LCD with I2C Module: An LCD (liquid-crystal display) is a flat-panel optical device that utilizes liquid crystals and polarizers to display information. When paired with an I2C (Inter-Integrated Circuit) module, the LCD can communicate with microcontrollers using the I2C protocol, simplifying connectivity and reducing wiring complexity.

Arduino Uno Board : The Arduino Uno is an open-source microcontroller board based on the microchip ATmega328P microcontroller and developed by Arduino.cc . The board is equipped

with sets of digital and analog input/output (I/O) pins that may be interfaced to various boards and other circuits Circuit Diagram.

DC Fan: The DC fan is the device you want to control. It typically has two wires for power supply - positive (+) and negative (-).

Relay: A relay is an electromechanical switch that can be controlled by the Arduino. It allows the Arduino to control high-power devices (like the DC fan) using low-power signals.

Flame Sensor: The flame sensor employed in our system boasts several key features essential for fire detection applications. It includes indicator lights, with a green LED indicating the switch status and a red LED indicating power. Additionally, the sensor incorporates a potentiometer for sensitivity control, allowing users to adjust the detection threshold based on environmental conditions. Its detection range spans from 760 to 1100 nm, effectively capturing flames and specific wavelengths associated with fire. With a detection angle of approximately 60 degrees, the sensor exhibits high sensitivity to flame spectra, making it suitable for fire alarm systems and safety applications.

Electromagnetic Buzzer: We integrated a 5V electromagnetic buzzer into our system to provide audible alerts in response to detected flame signals. Operating on a 5V power supply, the buzzer utilizes a coil element to generate an audible tone, ensuring clear and distinct notifications when activated. Its electromagnetic design offers reliable performance and robustness, making it suitable for various alerting and notification applications.

Micro Submersible Water Pump: The DC 3V mini Micro Submersible Water Pump serves as a vital component in our environmental monitoring and safety system. Designed for low-cost and small-size applications, the pump operates on a 2.5 ~ 5V power supply, offering versatility and ease of integration. With its submersible design, the pump can be submerged in water for efficient pumping operations. Its compact size and simple operation make it ideal for a wide range of applications, including irrigation systems, water circulation, and liquid transfer tasks within greenhouse environments

5 Design Analysis and Evaluation

5.1 Novelty

The novelty of our project lies in its integration of advanced sensor technologies and IoT analytics platforms to create an intelligent environmental monitoring and safety system tailored for greenhouse agriculture. By combining components such as the ESP8266 module, DHT11 sensor, LCD display, and Arduino Uno board, our system enables real-time monitoring of temperature, humidity, soil moisture, and fire detection. Leveraging ThingSpeak, an IoT analytics platform, facilitates data aggregation, visualization, and analysis in the cloud, enhancing efficiency and enabling proactive decision-making. Additionally, the incorporation of flame sensors, electromagnetic buzzers, and submersible water pumps adds an extra layer of

safety and automation, ensuring optimal conditions for plant growth while mitigating fire risks. Overall, our project represents an innovative approach to sustainable greenhouse management, addressing key challenges in environmental monitoring and safety with cutting-edge technology and comprehensive functionality.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

Our project prioritizes public safety by ensuring the accuracy and reliability of circuitry to prevent potential electrical hazards. Measures are implemented to minimize risks associated with electromagnetic radiation and ensure data security.

5.2.2 Considerations to environment

Environmental considerations include minimizing electronic waste generation, optimizing energy consumption through component selection, and promoting recycling and proper disposal practices. By integrating eco-friendly measures, we aim to reduce our project's carbon footprint and contribute to environmental sustainability.

5.3 Investigations (PO(d))

5.3.1 Results and Analysis

The algorithm of our code is:

Read temperature, humidity, and soil moisture from sensors

Display temperature, humidity, and soil moisture on LCD

Print temperature, humidity, and soil moisture to serial monitor

Check if flame is detected

 If flame is detected:

 Turn on fire pump and buzzer

 While flame is detected:

 Check if flame is still present

 Turn off fire pump and buzzer

Else:

Control light based on the following conditions:

If temperature < MIN_TEMPERATURE and LDR value < LDR_THRESHOLD:

Turn on light

Else if temperature > MAX_TEMPERATURE:

Turn off light

Else if soil moisture > 65 and temperature is between MIN_TEMPERATURE and MAX_TEMPERATURE:

Turn on light

Else if soil moisture is between OPTIMAL_MOISTURE_MIN and OPTIMAL_MOISTURE_MAX:

Turn off light

Else if humidity > MAX_HUMIDITY:

Turn on light

Else:

Turn off light

Control soil pump based on the following conditions:

If soil moisture < 20:

Turn on soil pump

While soil moisture < 50:

Read soil moisture

Turn off soil pump

Else if soil moisture > 65:

Turn on fan and light

While soil moisture > 60:

Read soil moisture and temperature

If temperature >= 24:

Turn off light

Turn off fan

Else if soil moisture is between 20 and 60:

Do nothing

Update ThingSpeak with temperature, humidity, and soil moisture data

5.3.2 Interpretation and Conclusions on Data

Our project was implemented step by step through several stages.

Firstly, we have built the temperature and humidity sensor. We use DHT11 Digital Relative Humidity & Temperature sensor Module here. Next we implement the LCD display to show our live temperature and humidity on the display. We have used 16*2 Serial LCD Module Display in this work. After implementing the whole thing, we compared it with Google Live data, it was showing almost exactly the same results.

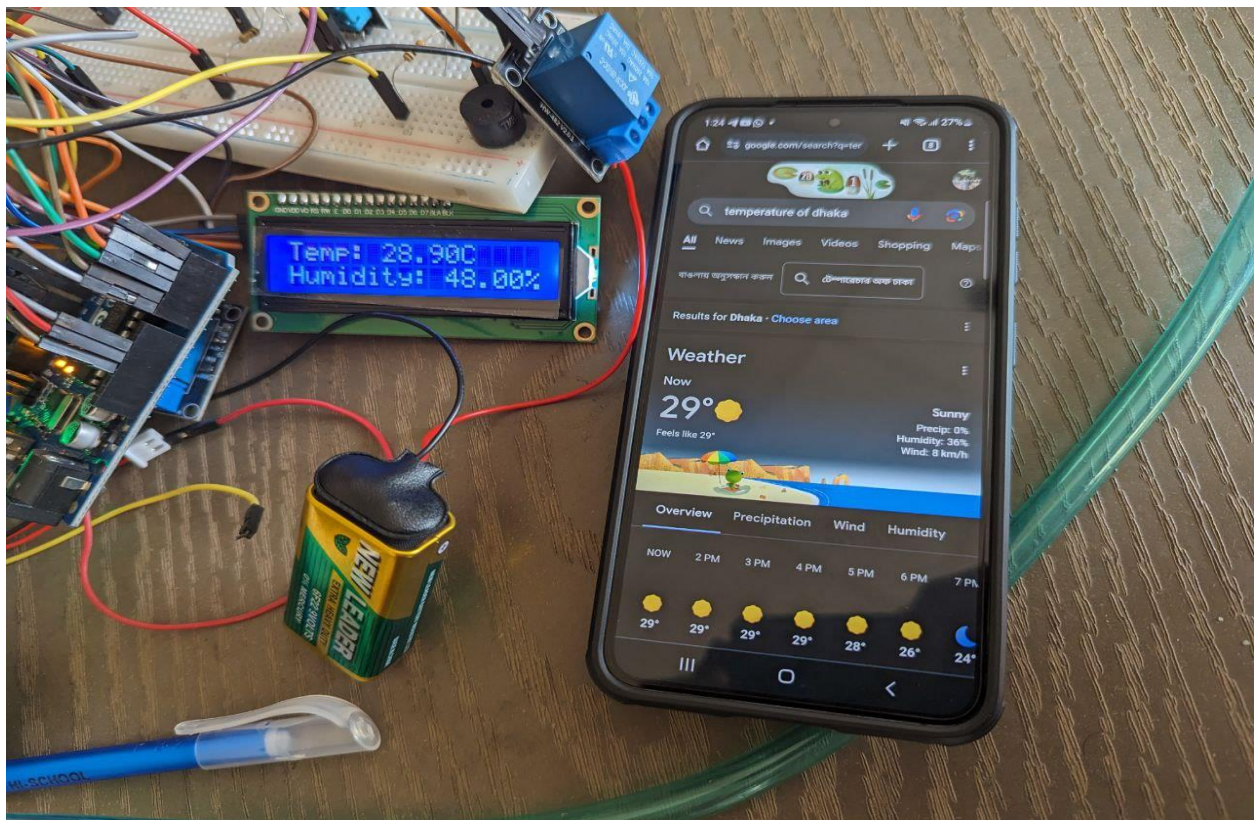
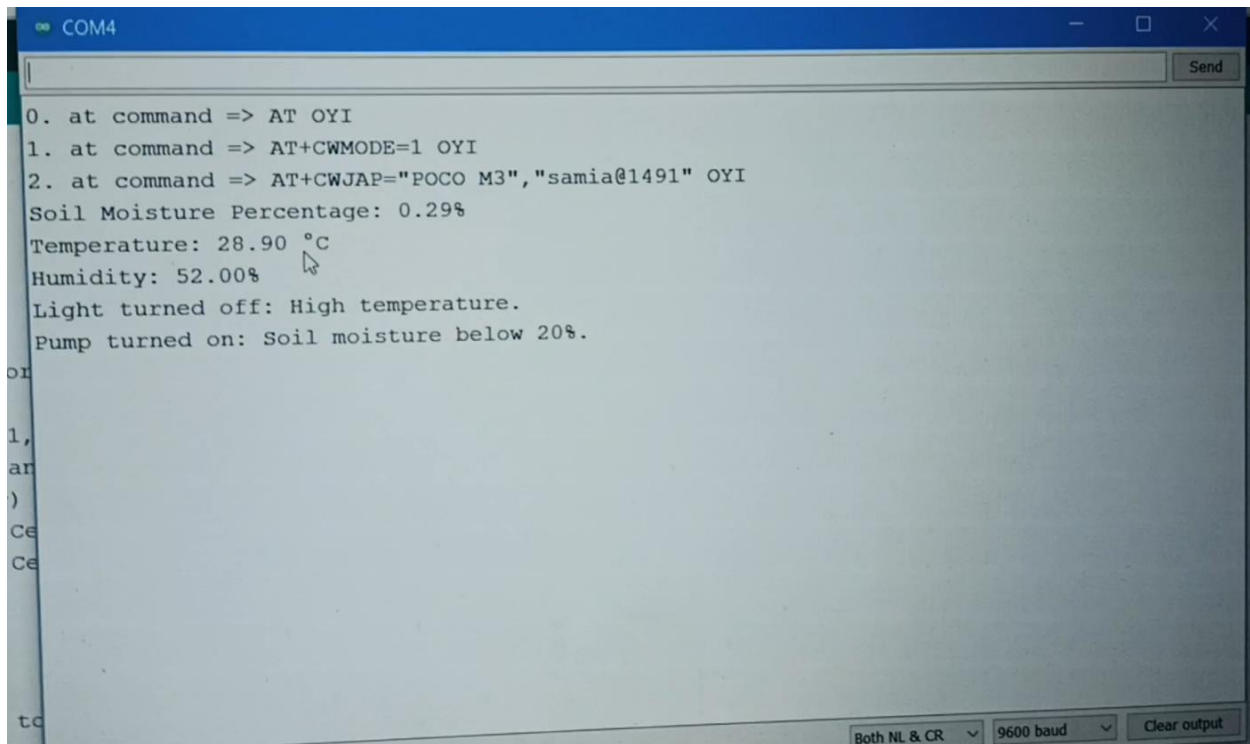


Fig: Live Temperature matching with our sensor reading

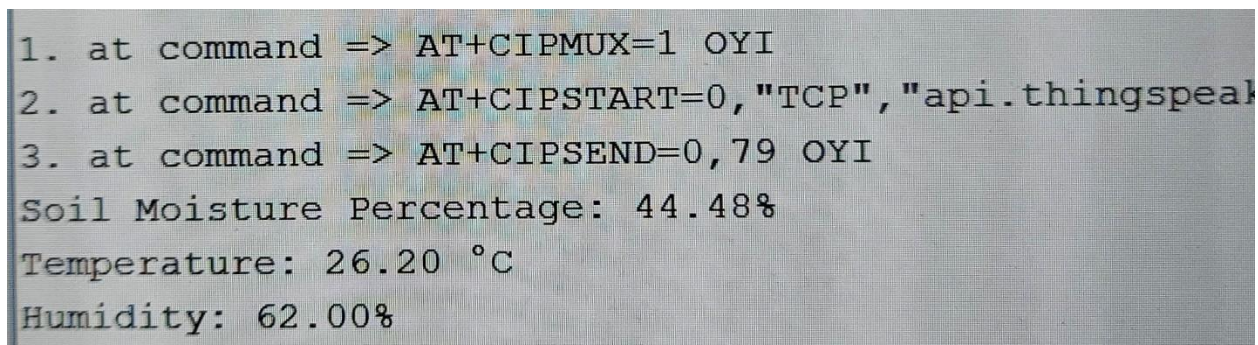
It was also showing at our Arduino output window



```
COM4
0. at command => AT OYI
1. at command => AT+CWMODE=1 OYI
2. at command => AT+CWJAP="POCO M3","samia@1491" OYI
Soil Moisture Percentage: 0.29%
Temperature: 28.90 °C
Humidity: 52.00%
Light turned off: High temperature.
Pump turned on: Soil moisture below 20%.
Both NL & CR 9600 baud Clear output
```

Fig: Live Temperature and humidity showing at output window

Then we implement the Soil Moisture and Humidity Sensor to measure the moisture(%) and humidity(%) of the soil. We use YL-69 Soil Hygrometer Humidity and Soil Moisture Detection sensor here. It was also showing almost exact result .



```
1. at command => AT+CIPMUX=1 OYI
2. at command => AT+CIPSTART=0,"TCP","api.thingspeak.com" OYI
3. at command => AT+CIPSEND=0,79 OYI
Soil Moisture Percentage: 44.48%
Temperature: 26.20 °C
Humidity: 62.00%
```

Fig: Live moisture percentage showing at output window

Then, we built the first pump with our circuit with a relay. We used 5v mini submersible pump, Relay Connection and a 9 V DC battery here. Then We made a condition by Arduino

Coding. We set it so that when the soil moisture sensor reading is below 20%, the pump will turn on and water the plants. This will continue until the moisture reading goes above 50%

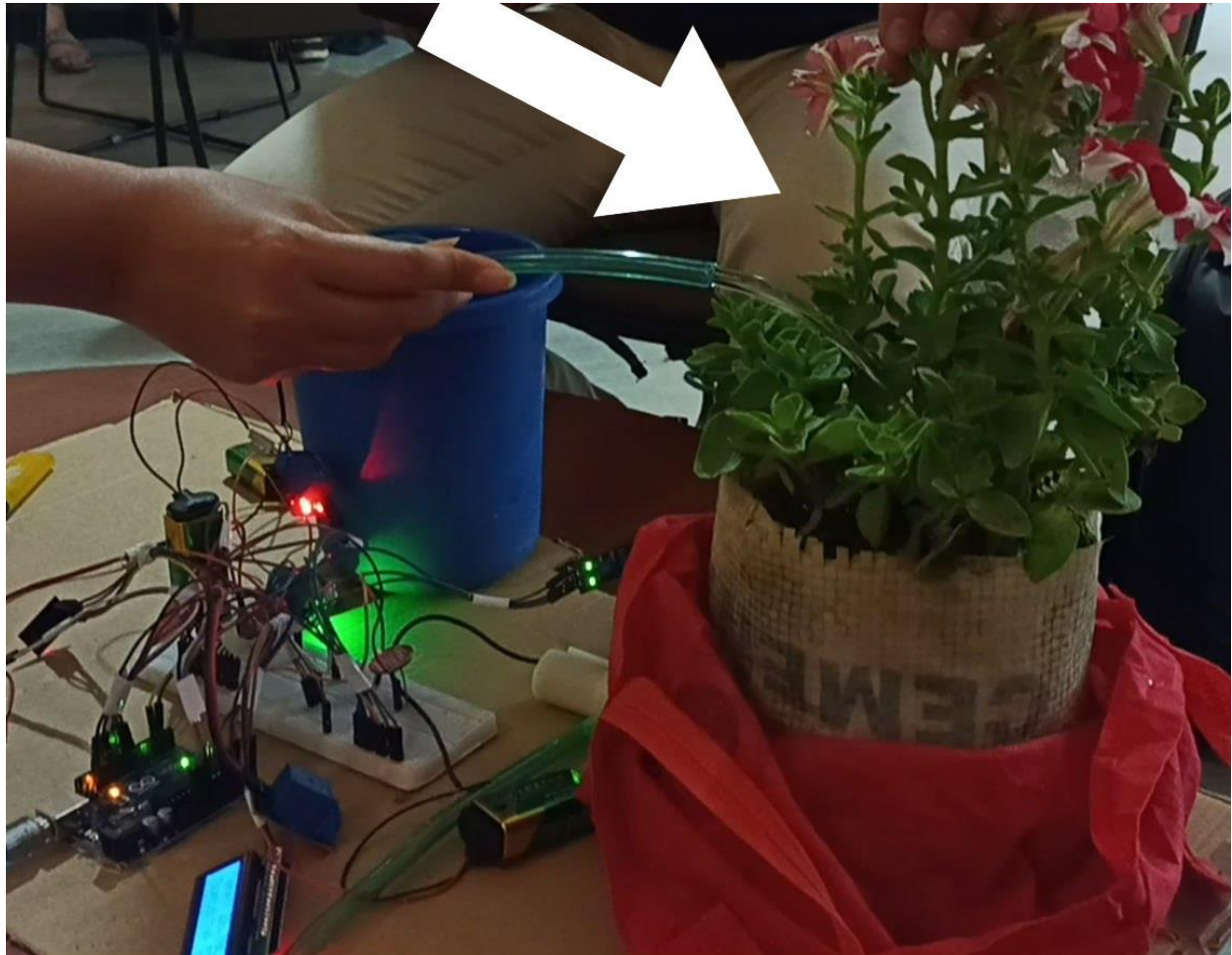


Fig: The Pump is pumping water after reading the moisture reading

Then We Built the fan connection with Relay Module . Again we set the condition in it such that when the soil moisture goes above 65% our fan automatically turns on.



Fig: The Fan is rotating according to our condition

We connect and build up an LDR and simultaneously an LED in the circuit and set the condition in such a way that when the humidity sensor is more than 65% the light will turn on. The LDR will work with the LED in such a way that when the temperature is high the resistance of the LDR decreases and then turns off the LED. The light will turn on again when the temperature is below 18°C, the soil moisture is above 65% and the humidity is above 80%.

Then we made the Flame sensor Portion. We used a Flame sensor Module to detect any fire or flame within a certain region. In this full system, the fire detection process will continue continuously, and if no fire is detected, the output display will show No Fire Detect.

```
Checking if there is fire!
No fire detected.
Light turned off: High temperature.
Checking if there is fire!
```

Fig: No Fire condition

But whenever our flame sensor can detect any fire around it will show fire detection on the display. We have added a buzzer as warning alarm in this system. As long as the sensor is in contact with the fire, the buzzer will continue to sound. As soon as the fire stops, the buzzer will stop sounding.

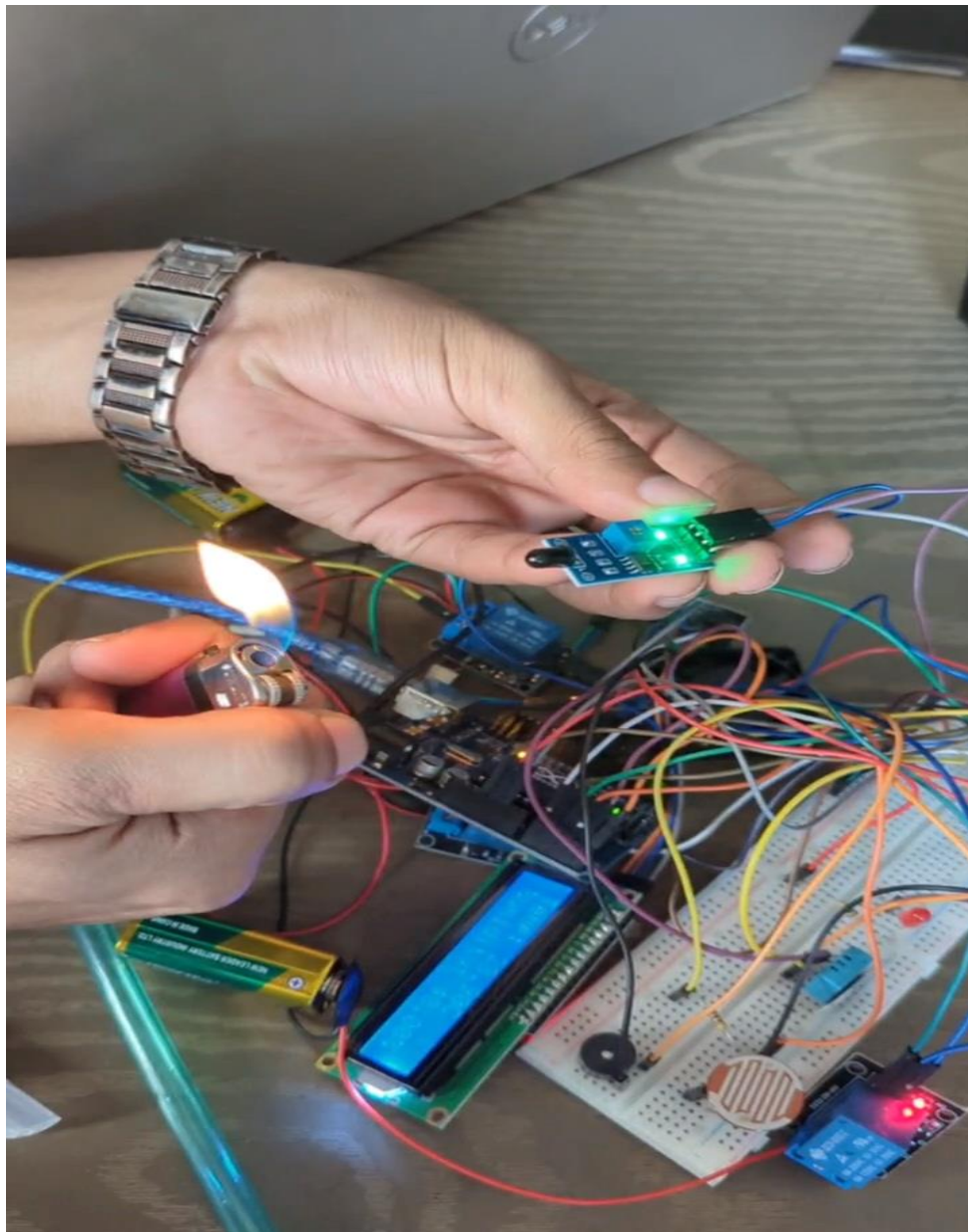


Fig: When our sensor was exposed to fire

At this stage we have connected our second pump. And its condition is set in such a way that as long as there is a fire, the pump will run to extinguish the fire, as soon as the fire goes out, the pump and water supply will stop.

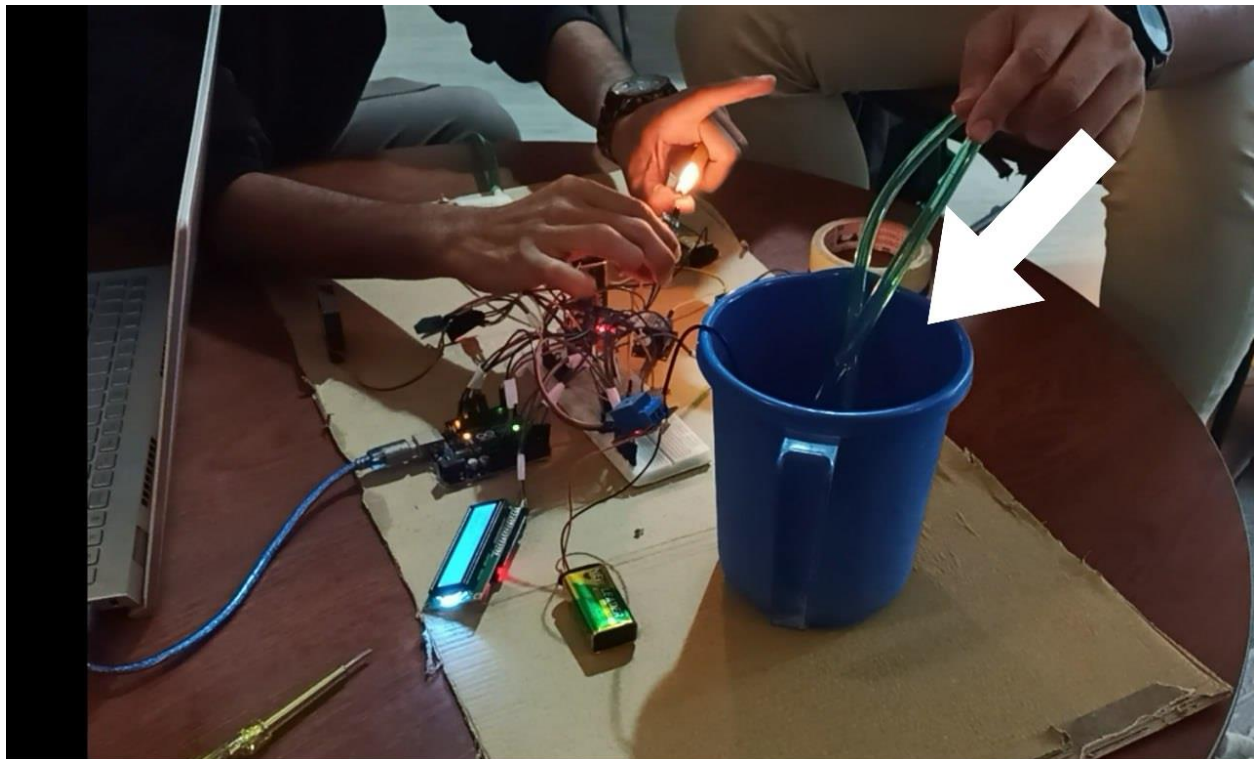


Fig: The fire is detected and the pump is flowing water

We used an ESP 8266 module in this full system so that all our data is stored in the cloud and we can store and analyze this data later.

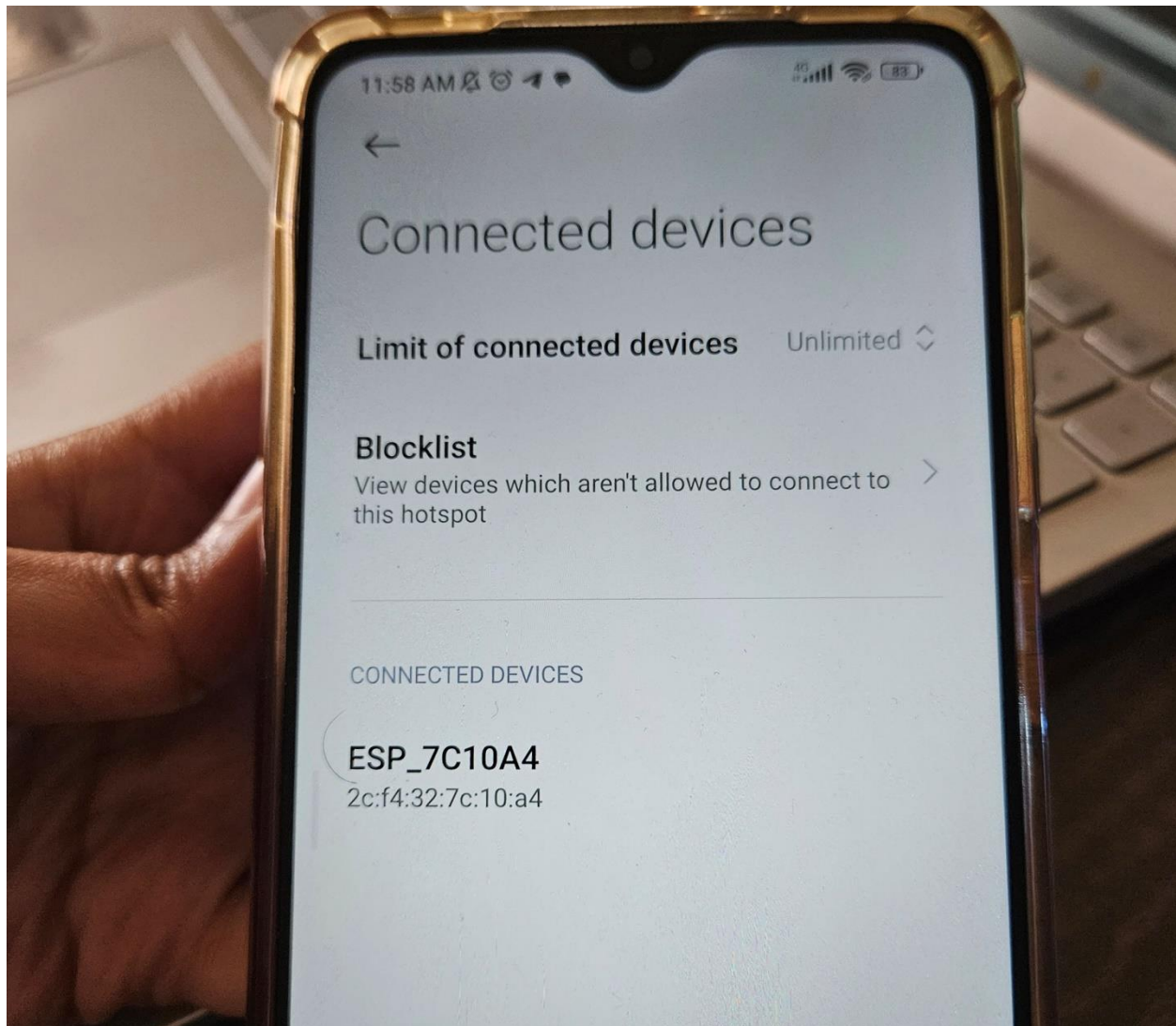


Fig: ESP8266 is connected with our system

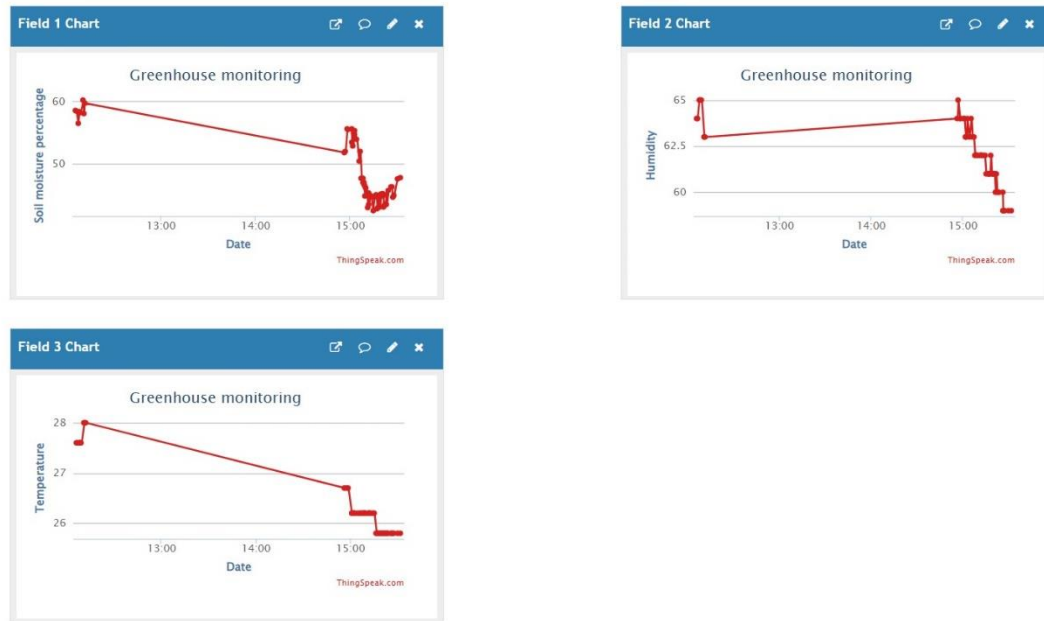


Fig: Data store and analysis in the cloud via ESP

5.4 Limitations of Tools (PO(e))

Fire sensors, while effective in detecting flames, may encounter challenges under specific conditions, such as bright sunlight or when obstructed by smoke or other objects. Proper placement and calibration of the fire sensor are critical for reliable detection in such scenarios.

Environmental factors like temperature, humidity, and dust can impact the performance of components over time, potentially leading to degradation or malfunction. Regular maintenance and protection from harsh conditions are necessary to mitigate these effects.

False alarms may occur occasionally due to factors such as dust, insects, or sudden temperature changes, affecting the system's reliability. Implementing algorithms or additional sensors to filter out false alarms can enhance the accuracy of the detection system.

Moreover, a notable limitation is the inability of flame sensors to precisely locate the flame's position, resulting in water sprinkling indiscriminately rather than targeting the fire source. Addressing this challenge requires innovative solutions to direct water accurately towards the flame, optimizing firefighting effectiveness while minimizing water wastage.

5.5 Impact Assessment (PO(f)):

5.5.1 Assessment of Societal and Cultural Issues:

Our project considers societal and cultural implications by addressing the pressing need for advanced agricultural technologies to meet growing food demands sustainably. By enhancing greenhouse management and fire safety systems, we contribute to agricultural sustainability and resilience, aligning with societal expectations for environmentally responsible practices.

5.5.2 Assessment of Health and Safety Issues:

Ensuring the health and safety of agricultural workers and farm environments is paramount. Our project's implementation of fire detection and prevention measures minimizes the risk of fire-related accidents in greenhouses, safeguarding both workers and crops. Additionally, by monitoring environmental parameters like temperature and humidity, we promote a safe and conducive working environment for agricultural operations.

5.5.3 Assessment of Legal Issues:

Compliance with relevant regulations and standards is essential to ensure the legality and legitimacy of our project. We adhere to fire safety regulations and building codes governing agricultural facilities, ensuring that our system meets the necessary legal requirements for deployment. Moreover, considerations of data privacy and security align with legal frameworks governing the collection and use of sensor data in agricultural settings.

5.6 Sustainability Evaluation (PO(g)):

Our project prioritizes environmental sustainability by optimizing resource utilization and minimizing environmental impact. Through efficient management of water resources using smart irrigation systems and proactive fire prevention measures, we reduce water wastage and mitigate the risk of environmental damage caused by fires. Furthermore, by promoting sustainable agricultural practices, our project contributes to long-term environmental resilience and ecosystem health.

5.7 Ethical Issues (PO(h)):

Ethical considerations are central to our project's design and implementation. We uphold principles of fairness, transparency, and accountability in all aspects of our work, ensuring equitable access to agricultural technologies and benefits for all stakeholders. Moreover, we prioritize ethical data collection and usage practices, respecting user privacy and confidentiality while leveraging data insights responsibly to improve agricultural productivity and sustainability.

6.Reflection on Individual and Teamwork (PO(i)):

6.1 Individual Contribution of Each Member

The team consisted of seven members, each contributing unique skills and expertise to the project:

Student ID	Contribution
2006138	Project planning, Circuit building, Major and maximum portion of the Coding
2006150	Buying components, circuit building, Circuit debugging, Coding
2006157	Buying components, circuit building, Circuit debugging, Coding
2006160	Circuit building, Circuit debugging, Coding

6.2 Mode of Team Work

The team employed a collaborative approach to maintain strong teamwork throughout the project, utilizing regular meetings to assign tasks, review progress, and address issues. Communication methods such as email, messaging apps, and video chats were utilized, especially for remote work. Responsibilities were allocated based on individual abilities and knowledge through defined roles and duties. Collaboration tools like Google Drive facilitated the sharing of resources and ensured efficient teamwork and version control. Emphasis was placed on providing constructive criticism and feedback to ensure quality outcomes and foster continuous improvement. Periodic updates were given during meetings to track project milestones and address any arising issues promptly.

6.3 Diversity Statement of Team

The team comprised individuals from various backgrounds, experiences, and perspectives, fostering a lively and diverse environment for collaboration. Each team member brought unique skills and expertise to the project, enhancing the overall effectiveness and creativity of the group. This diversity stimulated innovation and creativity, enabling a comprehensive exploration of the problem statement and the generation of viable solutions. Cultural diversity and differing viewpoints enriched discussions and decision-making, resulting in well-rounded conclusions and resolutions. Embracing diversity as a strength, the team leveraged the abilities and ideas of its members to overcome challenges and successfully accomplish the project.

6.4 Log Book of Project Implementation

Date	Milestone achieved	Individual Role	Team Role	Comments
10.01.24	Project Kickoff Meeting	Organized project kickoff meeting	Participated in planning	The team convened to discuss project goals, timelines, and roles.
15.01.24	Hardware Setup of DHT11, Soil sensor, ESP8266	2006150 & 2006157-bought necessary components - Everyone participated to build the circuit 2006138- Wrote code for sensor reading, actuator control, and data transmission	Assembling and configuring Arduino UNO board and sensors	Initial setup of hardware components
22.01.24	IoT Integration	Everyone participated to build the circuit	Establishing communication with Thingspeak Cloud	Enable data transfer to cloud platform
01.02.24	Connection of pumps and fan	Everyone participated to build the circuit	Implementing algorithms for temperature, humidity, and soil moisture control	Ensure optimal growing conditions for plants
15.02.24	Fire Detection System	Everyone participated to build the circuit	Design and integrate flame sensor and safety protocols	Enhance safety measures against fire outbreaks
22.02.24	Testing and Validation	2006138, 2006150, 2006157, 2006160- Debugged the circuit and wrote the code for the system	Quality Assurance	Verify system performance under various conditions
28.02.24	Documentation	Everyone contributed to the report writing	Compile project documentation and reports	Document project progress and outcomes

7. Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Embarking on an ambitious endeavor, "EcoWatch: Smart Greenhouse Environment Management with Fire Safety," is poised to redefine conventional approaches to greenhouse management and fire safety. Spearheaded by a team of dedicated innovators, the project is set to explore cutting-edge advancements in IoT, sensor technology, and automation. By integrating environmental monitoring with fire detection and safety protocols, EcoWatch aims to optimize resource utilization, enhance plant growth conditions, and mitigate the risks associated with fire outbreaks.

The project's focal points include:

- Implementing state-of-the-art sensor technology for real-time monitoring of temperature, humidity, and soil moisture levels.
- Developing intelligent algorithms to regulate environmental parameters and ensure optimal growing conditions.
- Integrating fire detection systems with proactive safety measures to detect and extinguish fires efficiently.
- Establishing seamless communication with cloud platforms for data logging, analysis, and remote monitoring.

Anticipated outcomes of the EcoWatch project include:

- Enhanced efficiency and sustainability in greenhouse environments.
- Improved safety measures through early detection and mitigation of fire hazards.
- Streamlined management of resources, leading to cost savings and increased productivity.

8. Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Cost Evaluation:			
Product	Unit price (BDT)	Quantity	Total price (BDT)
Battery 9V	75	6	450
5mm red LED	5	1	5

16x2 Serial LCD Module Display for Arduino Assembled	340	1	340
DHT11 Digital Relative Humidity and Temperature Sensor Module	150	2	300
Active Buzzer 5V	20	1	20
YL-69 Soil Hygrometer Humidity & Soil Moisture Detection Sensor	120	1	120
ESP8266 ESP-01 WIFI Wireless Transceiver Send Receive LWIP AP+STA M70	215	1	215
Standard LCD 16x2 Display	240	1	240
Flame Sensor Module	85	2	170
Mini Cooling Fan 40x40x10mm	150	1	150
5v Relay module for Arduino	75	3	225
1207 Passive Buzzer	15	1	15
I2C LCD Adapter Module	95	1	95

LDR 20 mm Light Depending Resistor	80	1	80
Submersible 3V Mini DC Water Pump with Wire Horizontal Type	160	2	320
Ohm (Ω) 1/4w Resistors - Pack of 5 - Resistor Pack of 5 : 100 Ω	5	1	5
Jumper Wire	50	3	150
Water Tube	25	1	25
Plant	80	1	80
Lighter	20	1	20

8.2 Timeline of Project Implementation



9. Future Work (PO(I))

The future work for our project embraces innovation and continuous improvement to address the complex challenges of environmental monitoring and safety systems.

Integrated Pest Management (IPM): we plan to bolster insect control measures within the greenhouse system. This involves integrating automated pest detection using sensors and image processing techniques to swiftly identify common greenhouse pests. Additionally, we aim to implement targeted pest management strategies, such as the application of biopesticides or natural predators, to mitigate harm to beneficial organisms. Smart trapping systems equipped with attractants and sensors will be deployed for effective monitoring and control of insect

populations. Furthermore, we will integrate insect control measures into the broader framework of Integrated Pest Management (IPM) for sustainable pest management. Developing data-driven predictive models for pest outbreaks and infestation risks, alongside exploring remote monitoring and control solutions using IoT devices and cloud platforms, will enable proactive intervention and optimal resource allocation. These advancements will not only safeguard crop health but also promote sustainability and resilience in greenhouse cultivation.

Dynamic Pump Control: One key aspect of future work involves enhancing the functionality of the pump by implementing dynamic control mechanisms. By integrating feedback from soil moisture sensors, the pump's speed can be adjusted dynamically to match the specific moisture requirements of different plant species. This ensures precise and efficient water delivery tailored to the individual needs of each plant.

Adaptive Fan Control: Future iterations of the system will focus on optimizing fan speed and direction based on real-time humidity and soil moisture data. By intelligently adjusting fan parameters to direct airflow towards areas with maximum humidity and soil moisture levels, the system can maintain optimal growing conditions while minimizing energy consumption.

Accurate Fire Sensing and Response: Enhancements in fire detection algorithms will enable the system to accurately identify and localize flames in the greenhouse. Once a fire is detected, the pump will be activated to supply water only to the affected area until the fire is extinguished, preventing water wastage and minimizing damage to surrounding plants.

Plant-Specific Customization: Recognizing that different plant species have varying environmental requirements, future work will focus on designing the system to accommodate the diverse needs of individual plants. By incorporating plant-specific parameters such as optimal temperature, humidity, and light levels, the system can dynamically adjust environmental conditions to ensure optimal growth and productivity for each plant species.

Integration of Advanced Control Algorithms: Advanced control algorithms, including machine learning and predictive analytics, will be integrated into the system to optimize resource allocation and decision-making processes. By analyzing historical data and environmental trends, the system can anticipate plant needs, predict potential issues, and proactively adjust system parameters to optimize performance and resilience.

User-Friendly Interface: Improvements in the user interface will enhance the system's accessibility and usability for greenhouse operators. Intuitive dashboards and interactive controls will enable users to monitor system status, adjust settings, and receive alerts or recommendations based on real-time data, facilitating informed decision-making and efficient management of greenhouse operations.

Enhanced Sensor Integration: Future iterations of the EcoWatch project will explore the integration of advanced sensor technologies to further expand the scope of environmental monitoring. This includes the incorporation of additional sensors for detecting air quality, nutrient levels, and plant health indicators, enabling comprehensive monitoring and optimization of greenhouse conditions.

Machine Learning and Predictive Analytics: Leveraging machine learning algorithms and predictive analytics, future work will aim to develop intelligent systems capable of forecasting environmental trends and anticipating potential hazards. By analyzing historical data and real-time sensor readings, these systems can provide proactive recommendations for optimizing resource utilization and preventing adverse events

By addressing these points in future work, the EcoWatch project aims to advance the state-of-the-art in greenhouse management and fire safety systems, providing sustainable solutions that optimize resource utilization, enhance plant health, and promote environmental resilience.

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